

An Immersive Multi-User VR-based System for the Training of Electrical Substation Maintenance

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ABSTRACT

Maintenance routines in electrical substations are considered as a high-risk engineering activity. In this context, systems based on Virtual Reality (VR) techniques are well explored for providing a safer and lower cost training environment. However, concurrent training is not supported in many tools developed for this purpose. Supporting such feature is very important to simulate real life maintenance routines. Thus, this work presents the development of a system for the utilization of immersive and multi-user VR-based techniques to support the concurrent training of electrical substations operators. To do so, we present an architecture that provides concurrent and immersive virtual training using head mounted displays such as Meta's Quest Pro. This involves designing and implementing a server-side architecture that can handle data storage and communication with clients.

Keywords: Multi-user, Virtual Reality, Training, Electrical substation.

Index Terms: Computing methodologies—Modeling and simulation—Simulation evaluation.

1 INTRODUCTION

Electrical substations, as fundamental and rudimentary components of electrical systems [1], require unceasing maintenance and vigilant oversight throughout their operational lifecycle. However, by virtue of their high-power capacity and voltage levels, the inherent hazards associated with their maintenance and operation are markedly significant [2]. Consequently, comprehensive training in substation operations and maintenance is imperative on a global scale. Nevertheless, such training is frequently conducted in an isolated environment, whereas in real life, operation is performed concurrently. This is primarily due to the innate perils involved. In many instances, conventional methodologies rely on document-centric pedagogy and furnish only superficial demonstrations of high-risk tasks [3], [4].

Despite the incorporation of simulators designed to replicate real-world conditions [1], these simulators, at their present state, are incapable of faithfully emulating the intricate dynamics and collaborative nature characteristic of authentic operational settings.

In the context of maintenance and operation of electrical substation training, Virtual Reality emerges as a prospective solution for mitigating risks. However, a notable challenge persists in the extant approach to VR-based training, which typically accommodates only a solitary trainee at a time [1], [4], [5].

Moreno et. al, [6], provided a multi-user system for maintenance of energy lines. However, this feature is not applicable in the training of electric components such as transformers, banks of capacitors. These intrinsic limitations impair the ability of trainers to discern and address emerging issues, thereby creating a meaningful gap between the training experience and genuine operational scenarios characterized by intricate teamwork and rigorous preparatory measures [7].

Therefore, with these training needs, there arises a requirement for the creation of a tool to build training scenarios closer to reality within the substation. This tool was developed utilizing Virtual Reality to enhance realism and closely simulate the actual operations conducted within substations. For the application demonstration, a training scenario was developed based on the operation of a capacitor bank, a clear example of teamwork, as measurements need to be taken in pairs.

Consequently, a multi-user environment was constructed, primarily aimed at training where multiple individuals can interact with the objects in the environment. The goal is to create an immersive experience that mirrors real-world scenarios and encourages collaborative interactions among users.

This interaction between operators and engineers within the substation is crucial, given that electrical substations experience a notable frequency of workplace accidents [5], despite the existence of statutory regulations and strong compliance requirements. This underscores the intrinsic dangers and highlights the urgent need for effective and realistic training programs aimed at reducing the frequency of accidents and, ideally, preventing them.

2 SYSTEM ARCHITECTURE

The multiplayer training system developed in this work is based on the architecture presented in Figure 1. The architecture is based on a server containing the data for each model where the procedure for substation maintenance must take place. As a case study and following company collaborators, the object of study is a bank of

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capacitors. A historical dataset from a supervisory system called SAGE is used to perform simulation of real-life operations.

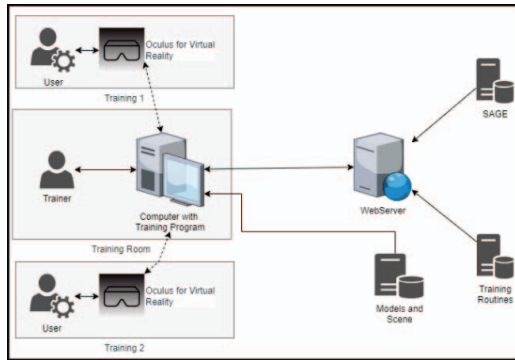


Figure 1: Proposed system architecture.

In this online server, training routines are stored. The server sends the data to a computer that will run the simulation, along with the models and the scene on that computer, which can be modified with the trainer's approval. Finally, this data is sent to the Oculus, running the VR simulation, which will be used by various trainees.

3 RESULTS

The system operates within an environment entirely modelled through parametric design within Unity 3D, facilitating scene creation by integrating both VR and computer functionality into a unified platform. Additionally, this environment serves as the very space where the operation of these equipment takes place, as illustrated in Figure 2.



Figure 2: Demonstration of the Real and the Virtual environment.

Therefore, the sense of strangeness in a virtual environment can be minimized. Subsequently, within this environment, the training is designed to allow the trainee to navigate through the space and view the equipment in true-to-life scale. When approaching the equipment, virtual tools, including two standard pliers and one circular plier, are generated for training, along with a model of the equipment commonly used in substations, the CB2000. With these tools, the user can initiate the scene, placing the pliers inside the capacitors to carry out measurements.

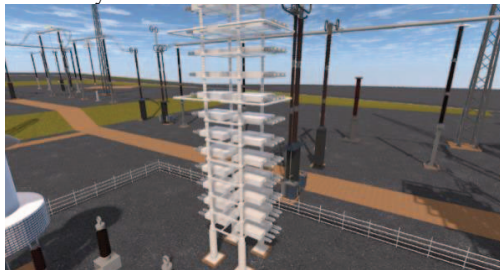


Figure 3: Bank of capacitors for training.

And since the training is conducted in a multi-user format, all participants in the scene can interact with the equipment and use

them to take measurements, as seen in Figure 4. This training format is crucial because training is always conducted as a team. Therefore, the need to collaborate and ensure safety becomes a point of utmost importance, unlike conventional individual training.

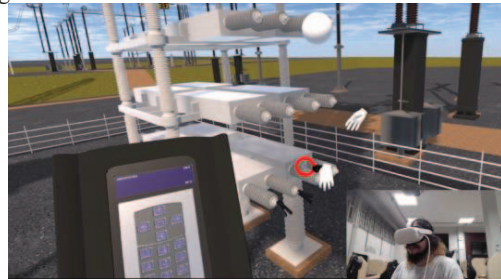


Figure 4: Virtual training with two operators in scene.

Therefore, this demonstrates a method for conducting multi-user training within a substation.

4 CONCLUSIONS

This paper aimed to demonstrate the feasibility of creating a VR multi-user platform to support concurrent engineering in training. It is believed that this approach can contribute to reduce risks during maintenance procedures in electrical substations.

ACKNOWLEDGMENTS

The authors would like to thank Eletronorte Eletrobras and all the support provided by the R&D (Research and Development) ANEEL - contract 4500062446. The authors also thank the Coordination for the Improvement of Higher Education Personnel (CAPES), CNPq, FAPEMIG for the financial support.

REFERENCES

- [1] I. F. Mondragón Bernal et al., "An Immersive Virtual Reality Training Game for Power Substations Evaluated in Terms of Usability and Engagement," *Applied Sciences*, vol. 12, no. 2, Art. no. 2, Jan. 2022, doi: 10.3390/app12020711.
- [2] C. S. V. Zuydam, M. S., and M. F., "Hazard and Risk Assessment in Electricity Sector: A Case of Swaziland Electricity Company," *J Geogr Nat Disast*, vol. s6, 2016, doi: 10.4172/2167-0587.S6-006.
- [3] Z. Chang, Y. Fang, Y. Zhang, and C. Hu, "A Training Simulation System for Substation Equipments Maintenance," in *2010 International Conference on Machine Vision and Human-machine Interface*, Apr. 2010, pp. 572–575. doi: 10.1109/MVHI.2010.31.
- [4] G. Lambert-Torres, G. M. Ribeiro, C. I. A. Costa, A. P. Alves da Silva, and V. H. Quintana, "Knowledge engineering tool for training power-substation operators," *IEEE Transactions on Power Delivery*, vol. 12, no. 2, pp. 694–699, Apr. 1997, doi: 10.1109/61.584348.
- [5] M. Fanqi and K. Yunqi, "An improved virtual reality engine for substation simulation," in *2010 2nd International Conference on Future Computer and Communication*, May 2010, pp. V1-846-V1-849. doi: 10.1109/ICFCC.2010.5497308.
- [6] Moreno, Erika F., et al. "Multi-user expert system for operation and maintenance in energized lines." In *2020 Future of Information and Communication Conference (FICC)*, Volume 2, doi: 10.1007/978-3-030-39442-4_34
- [7] J. I. Zambrano, D. A. Bermeo, C. A. Naranjo, and V. H. Andaluz, "Multi-User Virtual System for Training of the Production and Bottling Process of Soft Drinks," in *2020 15th Iberian Conference on Information Systems and Technologies (CISTI)*, Jun. 2020, pp. 1–7. doi: 10.23919/CISTI49556.2020.9141140.